Two-time descent: bicomplex identities and a $\mathbb{Z}_2$ obstruction (rigorous translation)

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(Dated: March 30, 2026)

CONVENTIONS

All sets and maps are in the ordinary category of sets. All groupoids are small. Composition of morphisms is written left-to-right: for morphisms $r \xrightarrow{p} s \xrightarrow{q} t$ we write $p \triangleright q : r \to t$ to mean "do p then q", i.e. $q \circ p$ in the usual right-to-left convention.

ASSUMPTIONS (STRUCTURAL DATA)

We fix the following abstract data.

(A1) Core SK data. A set of SK states SK together with a set of record states Sys and a map

$$\mathrm{diag} : \mathsf{Sys} \to \mathsf{SK}.$$

There is an involution (the SK swap)

$$\mathrm{swap} : \mathsf{SK} \to \mathsf{SK}, \qquad \mathrm{swap} \circ \mathrm{swap} = \mathrm{id}_{\mathsf{SK}}, \qquad \mathrm{swap} \circ \mathrm{diag} = \mathrm{diag}\,.$$

(A2) Two arrows (time and RG). Let T be a group with identity e_T, and set $\mathbb{N} := \mathbb{N}$ with identity 0 and composition $+$. We assume actions

$$\mathrm{act}_t : \mathsf{T} \times \mathsf{SK} \to \mathsf{SK}, \qquad \mathrm{act}_u : \mathbb{N} \times \mathsf{SK} \to \mathsf{SK}$$

satisfying the action laws

$$\mathrm{act}_t(e_T, x) = x, \qquad \mathrm{act}_t(t_2 t_1, x) = \mathrm{act}_t(t_2, \mathrm{act}_t(t_1, x)), \qquad \mathrm{act}_t(t^{-1}, \mathrm{act}_t(t, x)) = x,$$

$$\mathrm{act}_u(0, x) = x, \qquad \mathrm{act}_u(u_2 + u_1, x) = \mathrm{act}_u(u_2, \mathrm{act}_u(u_1, x)).$$

Finally, weak commutativity holds on SK:

$$\mathrm{act}_u(u, \mathrm{act}_t(t, x)) = \mathrm{act}_t(t, \mathrm{act}_u(u, x)).$$

(A3) Strict diagonalization of RG (record projection). For each $u \in \mathbb{N}$ with $u \geq 1$ there exists a map

$$\Pi_u : \mathsf{SK} \to \mathsf{Sys}$$

such that for all $x \in \mathsf{SK}$,

$$\mathrm{act}_u(u, x) = \mathrm{diag}(\Pi_u(x)).$$

We also assume a record-level RG action

$$R_u : \mathsf{Sys} \to \mathsf{Sys}$$

with monoid laws

$$R_0(r) = r, \qquad R_{u_2+u_1}(r) = R_{u_2}(R_{u_1}(r)),$$

and intertwining on diagonal states:

$$\Pi_u(\mathrm{diag}\, r) = R_u(r) \quad (u \geq 1),$$

equivalently,

$$\mathrm{act}_u(u, \mathrm{diag}\, r) = \mathrm{diag}(R_u(r)) \quad (u \geq 1).$$

(A4) RG-sector reachability (chosen scale). Fix $u_\star = 1$ and assume a section

$$\mathrm{sec}_\Pi : \mathsf{Sys} \to \mathsf{SK}$$

such that

$$\Pi_1(\mathrm{sec}_\Pi(r)) = r.$$

(A5) No mixed anomaly (bisimplicial descent laws). There is a bisimplicial set $\mathsf{C}_{m,n}$ (for $m, n \geq 0$) equipped with horizontal face maps $d_i^H : \mathsf{C}_{m+1,n} \to \mathsf{C}_{m,n}$ $(0 \leq i \leq m+1)$ and vertical face maps $d_j^V : \mathsf{C}_{m,n+1} \to \mathsf{C}_{m,n}$ $(0 \leq j \leq n+1)$ satisfying:

- **(A5.a)** $d_i^H d_j^H = d_{j-1}^H d_i^H$ for $i < j$;
- **(A5.b)** $d_i^V d_j^V = d_{j-1}^V d_i^V$ for $i < j$;
- **(A5.c)** $d_i^H d_j^V = d_j^V d_i^H$ for all i, j.

(A6) Record reflection (diagonal cancellation). If two diagonal SK states agree, then the underlying records agree:

$$\mathrm{diag}(r) = \mathrm{diag}(r') \Rightarrow r = r'.$$

Equivalently, the diagonal encoding of records has no redundancy.

Remark .1. Assumption **(A5)** constrains only *boundaries* (face maps) and thus ensures the two-time square has a well-defined boundary. Assumption **(A6)** is only a cancellation principle; record-level interchange is derived later from **(A2)**, **(A3)**, **(A4)**, and **(A6)** (Lemma .7). The "before/after" distinction is not encoded as two different equalities in Sys (UIP may collapse them); it is encoded in the process presentation by a binary tag $f \in \{\mathrm{diag}, \mathrm{bridge}\}$ on square generators. The meaning of this tag is fixed canonically by the Diff involution: diag acts as the identity and bridge acts as flip (Lemma .12). Nontriviality of this tag is established later once a non-fixed point of flip is obtained (Corollary .4), which is a consequence of the detector.

(A7) Real structure, universal pointer basis, and two-time mixing (analytic, used to derive a detector). There exists a complex Hilbert space $\mathcal{H}$ with an orthonormal basis $\{|a\rangle\}_{a\in\mathcal{I}}$ (the *pointer basis*), and we identify SK with a specified subset of trace-class operators on $\mathcal{H}$.

Universality of the pointer basis (record algebra is fixed). Let Diag denote the set of trace-class operators that are diagonal in the pointer basis. We assume that the record embedding is diagonal in this basis:

$$\mathrm{diag}(\mathsf{Sys}) \subseteq \mathsf{Diag}.$$

Equivalently, for each $r \in \mathsf{Sys}$ there exist coefficients $\{p_a(r)\}_{a\in\mathcal{I}}$ such that $\mathrm{diag}(r) = \sum_a p_a(r)|a\rangle\langle a|$. Moreover, all RG-diagonalized outputs land in the same diagonal subspace:

$$\mathrm{act}_u(u,x) \in \mathsf{Diag} \qquad (\forall x \in \mathsf{SK},\ \forall u \geq 1).$$

This packages the scale-stability of records: the "record basis" does not depend on when (or at which RG depth) diagonalization is applied [1–3].

Let $\mathcal{K}$ denote the antiunitary complex conjugation in this pointer basis.

In this concrete instantiation we take $\mathsf{T} = \mathbb{R}$ (additive group) and use microscopic unitary evolution only to derive the detector witness. The abstract act_t in the assumptions is the effective evolution on RG-classical states and is not assumed to preserve diagonality microscopically.

We assume:

(A7.a) (Swap as transpose/conjugation) the swap map is implemented by

$$\mathrm{swap}(\rho) = \mathcal{K}\,\rho^*\,\mathcal{K}^{-1},$$

so that in the pointer basis $\mathrm{swap}(\rho) = \rho^{\mathsf{T}}$.

(A7.b) (Time evolution) there is a nonzero bounded self-adjoint operator H such that

$$\rho(t) = e^{-itH}\,\rho\,e^{itH}.$$

(A7.c) (Scale/RG mixing generator; weakmini core) there exists a dense subspace $\mathcal{D} \subseteq \mathcal{H}$ such that $\mathcal{D} \subseteq \mathrm{Dom}(D)$, $H\mathcal{D} \subseteq \mathcal{D}$, $D\mathcal{D} \subseteq \mathcal{D}$, $K\mathcal{D} = \mathcal{D}$, and $KDK^{-1} = D$ on $\mathcal{D}$. The commutator is defined on $\mathcal{D}$ and satisfies

$$(DH - HD)\psi = iH\psi \qquad (\forall \psi \in \mathcal{D}). \tag{1}$$

(A7.d) (Nondegeneracy) there exist indices $a \neq b$ and a record state $r \in \mathsf{Sys}$ whose diagonal density matrix $\rho := \mathrm{diag}(r) = \sum_c p_c|c\rangle\langle c|$ satisfies $p_a \neq p_b$, and such that

$$\Re(H_{ab}) \neq 0. \tag{2}$$

Remark .2. Assumption **(A7)** is the physics-motivated bridge used to *derive* the swap-odd detector required by the obstruction argument, rather than postulating it. The purely combinatorial parts (bicomplex, ω-cover, splitting obstruction) use only **(A1)**–**(A6)**.

NO REAL SUBGROUP LEMMA AND DERIVED DETECTOR

Abstract algebraic lemma (formal core)

Definition .1 (Abstract mixing interface)**.** Let $(G, +, 0)$ be an abelian group. Fix endomorphisms

$$\mathrm{i} : G \to G, \qquad K : G \to G, \qquad \mathrm{ad}_D : G \to G.$$

Assume the identities

$$K(\mathrm{i}(x)) = \mathrm{i}(-K(x)), \qquad \text{(anti-linearity)} \tag{3}$$
$$K(\mathrm{ad}_D(x)) = \mathrm{ad}_D(K(x)), \qquad (K\text{-equivariance}) \tag{4}$$
$$\mathrm{ad}_D(-x) = -\mathrm{ad}_D(x), \qquad \text{(oddness)} \tag{5}$$

and the cancellation properties

$$x = -x \Rightarrow x = 0, \qquad \text{(no 2-torsion)} \tag{6}$$
$$\mathrm{i}(x) = 0 \Rightarrow x = 0. \qquad \text{(trivial kernel of i)} \tag{7}$$

An element $H \in G$ satisfies the *mixing relation* if $\mathrm{ad}_D(H) = \mathrm{i}(H)$. It is *$K$-odd* if $K(H) = -H$.

Remark .3 (Domain hygiene for the analytic instantiation)**.** In the Hilbert-space setting, we apply Definition .1 to the additive group of bounded operators that preserve $\mathcal{D}$ and for which the commutator with D is defined on $\mathcal{D}$. Under **(A7.c)**, $\mathrm{ad}_D(H)$ is defined on $\mathcal{D}$ and equals iH there.

Lemma .1 (No real subgroup)**.** *Under Definition .1, if an element $H \in G$ satisfies both*

$$\mathrm{ad}_D(H) = \mathrm{i}(H) \qquad \textit{and} \qquad K(H) = -H,$$

then $H = 0$.

Proof. Using K-equivariance and the mixing relation,

$$\mathrm{ad}_D(KH) = K(\mathrm{ad}_D H) = K(\mathrm{i}H) = \mathrm{i}(-KH).$$

Substitute $KH = -H$ to obtain $\mathrm{ad}_D(-H) = \mathrm{i}(-(-H)) = \mathrm{i}(H)$. By oddness, $\mathrm{ad}_D(-H) = -\mathrm{ad}_D(H)$, hence $-\mathrm{ad}_D(H) = \mathrm{i}(H)$. Replace $\mathrm{ad}_D(H)$ by $\mathrm{i}(H)$ to get $-\mathrm{i}(H) = \mathrm{i}(H)$, i.e. $\mathrm{i}(H) = -\mathrm{i}(H)$. No 2-torsion gives $\mathrm{i}(H) = 0$, and the trivial kernel of i gives $H = 0$. □

Remark .4. Lemma .1 is the abstract form of the statement that, under the weakmini core relation (1) on a K-invariant dense core and a K-even scale generator D on that core, one cannot have nontrivial dynamics entirely in the "real" subgroup characterized by $KHK^{-1} = -H$.

Derived detector in the pointer basis

Lemma .2 (Swap-odd observable constant on the diagonal)**.** *Fix distinct indices $a \neq b$ in the pointer basis and define the bounded self-adjoint operator*

$$O_{ab} := i\big(|a\rangle\langle b| - |b\rangle\langle a|\big).$$

Define

$$\mathrm{obs}_{ab}(\rho) := \mathrm{Tr}(\rho\, O_{ab}).$$

Then for every self-adjoint *trace-class ρ:*

- ***(i)*** *For every diagonal density matrix ρ in the pointer basis,* $\mathrm{obs}_{ab}(\rho) = 0$.
- ***(ii)*** $\mathrm{obs}_{ab}(\mathrm{swap}\,\rho) = -\,\mathrm{obs}_{ab}(\rho)$.
- ***(iii)*** *Moreover* $\mathrm{obs}_{ab}(\rho) = 2\,\Im(\rho_{ab})$ *where* $\rho_{ab} = \langle a|\rho|b\rangle$.

Proof. (i) If ρ is diagonal then $\rho_{ab} = 0$ for $a \neq b$, so $\mathrm{Tr}(\rho O_{ab}) = 0$.

(ii) In the pointer basis, $\mathrm{swap}(\rho) = \rho^{\mathsf{T}}$, hence $(\mathrm{swap}\,\rho)_{ab} = \rho_{ba} = \overline{\rho_{ab}}$ for self-adjoint ρ. Therefore $\Im((\mathrm{swap}\,\rho)_{ab}) = -\Im(\rho_{ab})$, and (iii) below implies (ii).

(iii) Directly compute

$$\mathrm{Tr}(\rho O_{ab}) = i(\rho_{ba} - \rho_{ab}) = 2\,\Im(\rho_{ab}),$$

using $\rho_{ba} = \overline{\rho_{ab}}$ for trace-class ρ when ρ is self-adjoint. □

Lemma .3 (Nontriviality of the derived detector)**.** *Under Assumption **(A7)**, there exist indices $a \neq b$ and a (time-evolved) state $\rho(t)$ such that*

$$\mathrm{obs}_{ab}(\rho(t)) \neq 0.$$

Consequently, the map $\mathrm{obs} := \mathrm{obs}_{ab}$ *is constant on diagonal record states and is swap-odd, while not being identically zero.*

Proof. Let $r \in \mathsf{Sys}$ and $\rho := \mathrm{diag}(r)$ be as in **(A7.d)** and consider its unitary time evolution $\rho(t) = e^{-itH}\rho e^{itH}$. Differentiate at $t = 0$:

$$\dot\rho(0) = -i[H, \rho].$$

The (a, b)-matrix element satisfies

$$\dot\rho_{ab}(0) = -iH_{ab}(p_b - p_a).$$

Write $H_{ab} = \Re(H_{ab}) + i\Im(H_{ab})$. Then

$$\dot\rho_{ab}(0) = (p_b - p_a)\Im(H_{ab}) - i(p_b - p_a)\Re(H_{ab}).$$

Taking imaginary parts gives

$$\frac{d}{dt}\Im(\rho_{ab}(t))\Big|_{t=0} = -(p_b - p_a)\Re(H_{ab}).$$

By (2), the right-hand side is nonzero, hence for sufficiently small t we have $\Im(\rho_{ab}(t)) \neq 0$, and by Lemma .2(iii) this implies $\mathrm{obs}_{ab}(\rho(t)) \neq 0$.

Finally, Lemma .2(i)–(ii) shows that obs_{ab} is constant on diagonal record states and is swap-odd. □

Remark .5 (Where Lemma .1 enters)**.** The nondegeneracy condition (2) requires a matrix element with nonzero real part. The "real subgroup loophole" would allow choosing a pointer basis in which H is K-odd (hence purely imaginary off-diagonal entries and no real part). Lemma .1 excludes that loophole when the weakmini core relation (1) holds on a K-invariant dense core and $H \neq 0$. Thus, under mixing, some basis element must have $\Re(H_{ab}) \neq 0$ for some pair. *By the universality clause in **(A7)**, the diagonal record algebra is fixed across RG depth, so the indices a, b appearing here label the same record outcomes in any procedure we compare.*

TWO-TIME CHAINS AND BICOMPLEX

Free abelian chains

For each (m, n) set

$$C_{m,n} := \mathbb{Z}[\mathsf{C}_{m,n}],$$

the free abelian group on the set $\mathsf{C}_{m,n}$. We write $[x] \in C_{m,n}$ for the generator corresponding to $x \in \mathsf{C}_{m,n}$.

Each face map induces a homomorphism on chains by linear extension, still denoted d_i^H, d_j^V, so $d_i^H([x]) = [d_i^H x]$ and similarly for d_j^V.

Two boundaries

Define homomorphisms

$$Q_t : C_{m+1,n} \to C_{m,n}, \quad Q_t([x]) := \sum_{i=0}^{m+1} (-1)^i\, [d_i^H x], \tag{8}$$

$$Q_u : C_{m,n+1} \to C_{m,n}, \quad Q_u([y]) := \sum_{j=0}^{n+1} (-1)^{m+j}\, [d_j^V y]. \tag{9}$$

The factor $(-1)^m$ in Q_u is the Koszul twist by horizontal degree.

Lemma .4 (Boundary-of-boundary expansion)**.** *Let d_i be face maps of a simplicial object and define $\partial = \sum_i (-1)^i d_i$. Then on generators,*

$$\partial^2 = \sum_{0\le i<j\le n} (-1)^{i+j}\big(d_i d_j - d_{j-1} d_i\big).$$

Proof. Write

$$\partial^2 = \sum_{i,j} (-1)^{i+j} d_i d_j$$

and split the sum into $i < j$ and $i \ge j$. For the $i \ge j$ part, set $i' = i + 1$, so $i' > j$ and $(-1)^{i+j} = -(-1)^{i'+j}$, giving

$$\sum_{i\ge j} (-1)^{i+j} d_i d_j = -\sum_{j<i'} (-1)^{i'+j} d_{i'-1} d_j.$$

Rename (i', j) as (j, i) to obtain

$$\sum_{i\ge j} (-1)^{i+j} d_i d_j = -\sum_{i<j} (-1)^{i+j} d_{j-1} d_i,$$

and combine with the $i < j$ part to get the claimed formula. □

Theorem .1 (Bicomplex identities)**.** *For all $m, n \ge 0$ the maps satisfy*

$$Q_t^2 = 0, \qquad Q_u^2 = 0, \qquad Q_t Q_u + Q_u Q_t = 0.$$

Proof. We prove each identity on generators.

(1) $Q_t^2 = 0$. Let $x \in \mathsf{C}_{m+2,n}$. Then

$$Q_t^2([x]) = \sum_{0\le i<j\le m+2} (-1)^{i+j}\big[d_i^H d_j^H x - d_{j-1}^H d_i^H x\big]$$

by Lemma .4. The simplicial identity $d_i^H d_j^H = d_{j-1}^H d_i^H$ for $i < j$ makes every summand zero.

(2) $Q_u^2 = 0$. Let $y \in \mathsf{C}_{m,n+2}$. Then

$$Q_u^2([y]) = \sum_{0\le j<k\le n+2} (-1)^{j+k}\big[d_j^V d_k^V y - d_{k-1}^V d_j^V y\big],$$

again by Lemma .4; the vertical face law $d_j^V d_k^V = d_{k-1}^V d_j^V$ kills each summand.

(3) Anticommutation. Let $z \in \mathsf{C}_{m+1,n+1}$. Expand

$$Q_t Q_u([z]) = \sum_{j=0}^{n+1}\sum_{i=0}^{m+1} (-1)^{m+1+j+i}[d_i^H d_j^V z],$$

$$Q_u Q_t([z]) = \sum_{i=0}^{m+1}\sum_{j=0}^{n+1} (-1)^{m+j+i}[d_j^V d_i^H z].$$

By mixed commutation $d_i^H d_j^V = d_j^V d_i^H$, the terms match, while the coefficients differ by a global minus sign because $(-1)^{m+1+j+i} = -(-1)^{m+j+i}$. Thus $Q_t Q_u([z]) + Q_u Q_t([z]) = 0$. □

RECORD PROCESS GROUPOID AND THE PARITY COCYCLE

This section uses only Assumptions **(A1)**–**(A6)**.

Definition .2 (Record-level maps)**.** We write $R_u : \mathsf{Sys} \to \mathsf{Sys}$ for the record-level RG action from **(A3)**. The record-level time action is derived from the RG-sector section **(A4)**:

$$T_t(r) := \Pi_1(\mathrm{act}_t(t, \mathrm{sec}_\Pi(r))).$$

In particular, $R_0(r) = r$ and $R_{u_2+u_1}(r) = R_{u_2}(R_{u_1}(r))$, and for $u \ge 1$ we have

$$\mathrm{act}_u(u, \mathrm{diag}\, r) = \mathrm{diag}(R_u(r)).$$

Lemma .5 (Derived diagonal equivariance on the RG sector)**.** *For all $t \in \mathsf{T}$ and $r \in \mathsf{Sys}$,*

$$\mathrm{act}_t(t, \mathrm{diag}\, r) = \mathrm{diag}(T_t(r)).$$

Proof. By **(A4)** and diagonalization, $\mathrm{diag}\, r = \mathrm{act}_u(1, \mathrm{sec}_\Pi(r))$. Apply weak commutativity and the factorization of $\mathrm{act}_u(1, \cdot)$ through diag to obtain the claim. □

Lemma .6 (Derived record-time action laws)**.** *For all $r \in \mathsf{Sys}$ and $t, t_1, t_2 \in \mathsf{T}$,*

$$T_{e_T}(r) = r, \qquad T_{t_2 t_1}(r) = T_{t_2}(T_{t_1}(r)), \qquad T_{t^{-1}}(T_t(r)) = r.$$

Proof. The identity law follows from $\mathrm{act}_t(e_T, x) = x$ and the section property $\Pi_1(\mathrm{sec}_\Pi(r)) = r$. For composition and inverse, observe that if $\Pi_1(x) = \Pi_1(y)$ then $\Pi_1(\mathrm{act}_t(t, x)) = \Pi_1(\mathrm{act}_t(t, y))$ by diagonalization, weak commutativity, and record reflection. Apply this with $x = \mathrm{act}_t(t_1, \mathrm{sec}_\Pi(r))$ and $y = \mathrm{sec}_\Pi(T_{t_1}(r))$, and similarly for inverses. □

Lemma .7 (Derived record-level interchange)**.** *Under Assumptions **(A2)**, **(A3)**, **(A4)**, and **(A6)**, for all $t \in \mathsf{T}$, $u \in \mathbb{N}$, and $r \in \mathsf{Sys}$,*

$$R_u(T_t(r)) = T_t(R_u(r)).$$

Proof. If $u = 0$, then $R_0 = \mathrm{id}$ and the claim is immediate. If $u \ge 1$, apply weak commutativity on $\mathrm{diag}(r)$, rewrite both sides using Lemma .5 and diagonalization, and cancel diag using record reflection (A6). □

Definition .3 (Typed syntax Proc_5)**.** For records $r, s \in \mathsf{Sys}$, define a set of morphisms $\mathrm{Proc}_5(r, s)$ inductively by the constructors:

(G1) $\mathrm{id}_r : \mathrm{Proc}_5(r, r)$.

(G2) $\mathrm{time}(t, r) : \mathrm{Proc}_5(r, T_t(r))$ for $t \in \mathsf{T}$.

(G3) $\mathrm{rg}(u, r) : \mathrm{Proc}_5(r, R_u(r))$ for $u \in \mathbb{N}$.

(G4) $\mathrm{sq}(f;t,u,r) : \mathrm{Proc}_5(R_u(T_t(r)), T_t(R_u(r)))$ for $f \in \{\mathrm{diag}, \mathrm{bridge}\}$.

(G5) Composition: if $p : \mathrm{Proc}_5(r,s)$ and $q : \mathrm{Proc}_5(s,t)$, then $p \triangleright q : \mathrm{Proc}_5(r,t)$.

(G6) Inverse: if $p : \mathrm{Proc}_5(r,s)$ then $p^{-1} : \mathrm{Proc}_5(s,r)$.

(G7) Codomain cast: if $p : \mathrm{Proc}_5(r,s)$ and $e : s = s'$, then $\mathrm{cast}_R^e(p) : \mathrm{Proc}_5(r,s')$.

Definition .4 (Presented equality $\approx_5$)**.** Let $\approx_5$ be the smallest relation on $\mathrm{Proc}_5(r,s)$ closed under:

(R1) Equivalence rules (refl/sym/trans) and congruence under composition, inverse, and cast.

(R2) Groupoid laws: associativity, left/right identity, inverse laws, and $(p^{-1})^{-1} = p$, plus

$$(p \triangleright q)^{-1} \approx_5 q^{-1} \triangleright p^{-1}, \qquad (\mathrm{id}_r)^{-1} \approx_5 \mathrm{id}_r.$$

(R3) Cast law: $\mathrm{cast}_R^{\mathrm{refl}}(p) \approx_5 p$.

(R4) Time laws (with explicit casts):

$$\mathrm{cast}_R^{T_{e_T}(r)=r}(\mathrm{time}(e_T, r)) \approx_5 \mathrm{id}_r,$$

$$\mathrm{cast}_R^{T_{t_2 t_1}(r)=T_{t_2}(T_{t_1}(r))}(\mathrm{time}(t_2 t_1, r)) \approx_5 \mathrm{time}(t_1, r) \triangleright \mathrm{time}(t_2, T_{t_1}(r)),$$

$$\mathrm{cast}_R^{T_{t^{-1}}(T_t(r))=r}\big(\mathrm{time}(t,r) \triangleright \mathrm{time}(t^{-1}, T_t(r))\big) \approx_5 \mathrm{id}_r.$$

(R5) RG laws (with explicit casts):

$$\mathrm{cast}_R^{R_0(r)=r}(\mathrm{rg}(0,r)) \approx_5 \mathrm{id}_r,$$

$$\mathrm{cast}_R^{R_{u_2+u_1}(r)=R_{u_2}(R_{u_1}(r))}(\mathrm{rg}(u_2+u_1, r)) \approx_5 \mathrm{rg}(u_1, r) \triangleright \mathrm{rg}(u_2, R_{u_1}(r)).$$

Crucially, there is ***no*** *relation equating a composite containing* $\mathrm{sq}(f;t,u,r)$ *with a composite that does not contain it.* In particular, the two fillers $\mathrm{sq}(\mathrm{diag};t,u,r)$ and $\mathrm{sq}(\mathrm{bridge};t,u,r)$ are not identified by any relation. The operational meaning of the tag is fixed canonically by Lemma .12 ($\mathrm{diag} \mapsto \mathrm{id}$, $\mathrm{bridge} \mapsto \mathrm{flip}$ on Diff). Nontriviality of the tag action is proved later from the detector (Corollary .4).

Definition .5 (Presented groupoid $\mathcal{G}$)**.** Objects are records $r \in \mathsf{Sys}$, and morphisms are $\mathrm{Proc}_5(r,s)$ modulo the relation $\approx_5$.

Definition .6 (Corner path and square-loop)**.** For $f \in \{\mathrm{diag}, \mathrm{bridge}\}$ define the canonical path to the "upper-right corner"

$$\mathrm{toCorner}_f(t,u,r) := \mathrm{time}(t,r) \triangleright \mathrm{rg}(u, T_t(r)) \triangleright \mathrm{sq}(f;t,u,r),$$

a morphism $r \to T_t(R_u(r))$. The *square-loop* is the loop at r defined by

$$\mathrm{sqLoop}(t,u,r) := \mathrm{toCorner}_{\mathrm{diag}}(t,u,r) \triangleright \mathrm{toCorner}_{\mathrm{bridge}}(t,u,r)^{-1}.$$

Definition .7 (Parity cocycle ω)**.** Define ω on generators by

$$\omega(\mathrm{time}) = 0, \qquad \omega(\mathrm{rg}) = 0, \qquad \omega(\mathrm{sq}(\mathrm{diag})) = 0, \qquad \omega(\mathrm{sq}(\mathrm{bridge}))$$

and extend by

$$\omega(p \triangleright q) = \omega(p) + \omega(q) \pmod 2, \qquad \omega(p^{-1}) = \omega(p), \qquad \omega(\mathrm{cast}_R^e(p))$$

Lemma .8 (ω is well-defined and detects the square-loop)**.** *The function ω respects all defining relations of $\mathcal{G}$ and thus defines a functor $\omega : \mathcal{G} \to B\mathbb{Z}_2$. Moreover,*

$$\omega(\mathrm{sqLoop}(t,u,r)) = 1 \qquad (t,u,r).$$

Proof. By definition, ω is additive under composition, invariant under inverses, and ignores casts, so it respects the groupoid and cast laws. It also respects the time and RG relations because those are generated entirely by the morphisms time and rg, on which ω is identically 0.

Crucially, the presentation of $\mathcal{G}$ includes *no* relation equating a composite that contains a square filler $\mathrm{sq}(f;t,u,r)$ with one that does not (Definition .3). Hence the assignments $\omega(\mathrm{sq}(\mathrm{diag})) = 0$ and $\omega(\mathrm{sq}(\mathrm{bridge})) = 1$ are consistent.

Finally, using Definition .6,

$$\omega(\mathrm{sqLoop}(t,u,r)) = \omega(\mathrm{toCorner}_{\mathrm{diag}}(t,u,r)) + \omega(\mathrm{toCorner}_{\mathrm{bridge}}(t,u,r$$

because ω vanishes on time and rg and differs only on the chosen filler tag. □

Corollary .1 (ω oddness on abelian coefficients)**.** *Let G be an abelian group and define an action by*

$$\mathrm{act}_\omega^G(p)(x) := (-1)^{\omega(p)} x.$$

Then for all t, u, r and $x \in G$,

$$\mathrm{act}_\omega^G(\mathrm{sqLoop}(t,u,r))(x) = -x.$$

Proof. By Lemma .8, $\omega(\mathrm{sqLoop}) = 1$, so $(-1)^{\omega(\mathrm{sqLoop})} = -1$. □

THE $\mathbb{Z}_2$-COVER AND SPLITTING OBSTRUCTION

This section uses only Assumptions **(A1)**–**(A6)**.

Definition .8 ($\mathbb{Z}_2$-cover of $\mathcal{G}$)**.** Objects are pairs (r,b) with $r \in \mathsf{Sys}$ and $b \in \mathbb{Z}_2$. A morphism $(r,b) \to (s,c)$ is a morphism $p : r \to s$ in $\mathcal{G}$ such that

$$c = b + \omega(p) \pmod 2.$$

Composition and inverses are inherited from $\mathcal{G}$. The projection functor $\pi : \widetilde{\mathcal{G}} \to \mathcal{G}$ forgets the parity.

Definition .9 (Splitting)**.** A *splitting* is a functor $s : \mathcal{G} \to \widetilde{\mathcal{G}}$ with $\pi \circ s = \mathrm{id}_{\mathcal{G}}$ [4].

Lemma .9 (Uniqueness of lifts in the $\mathbb{Z}_2$-cover)**.** *Fix objects (r, b) and (s, c) in $\widetilde{\mathcal{G}}$. For a given base morphism $p : r \to s$ in $\mathcal{G}$, there exists a morphism $(r, b) \to (s, c)$ lying over p if and only if*

$$c = b + \omega(p) \pmod 2.$$

When it exists, it is unique.

Proof. By definition of $\widetilde{\mathcal{G}}$, a morphism over p is not extra structure: it is exactly the datum of the base arrow p together with the parity constraint $c = b + \omega(p)$. Since p and (r, b) determine c uniquely, there can be at most one lift. Existence is exactly the satisfaction of the constraint. □

Lemma .10 (Splitting iff ω is a coboundary)**.** *There exists a splitting if and only if there exists a function $\sigma : \mathsf{Sys} \to \mathbb{Z}_2$ such that for every morphism $p : r \to s$ in $\mathcal{G}$,*

$$\omega(p) = \sigma(r) + \sigma(s) \pmod 2.$$

Proof. If s is a splitting, define $\sigma(r)$ by $s(r) = (r, \sigma(r))$. The lifted arrow $s(p)$ must satisfy $\sigma(s) = \sigma(r) + \omega(p)$ by the defining condition of $\widetilde{\mathcal{G}}$.

Conversely, given σ satisfying the displayed equation, lift each object to $(r, \sigma(r))$. For each $p : r \to s$ in $\mathcal{G}$, define $s(p)$ to be the unique lift $(r, \sigma(r)) \to (s, \sigma(s))$ lying over p; this lift exists by the parity equation $\sigma(s) = \sigma(r) + \omega(p)$ and is unique by Lemma .9. Additivity of ω implies functoriality. □

Corollary .2 (Odd loop obstructs splitting)**.** *If there exists a loop $p : r \to r$ with $\omega(p) = 1$, then no splitting exists.*

Proof. If a splitting existed, Lemma .10 would imply $\omega(p) = \sigma(r) + \sigma(r) = 0$, contradiction. □

Theorem .2 (Non-splitting of the $\mathbb{Z}_2$-cover)**.** *Under Assumptions **(A1)–(A6)**, if Sys is nonempty then the $\mathbb{Z}_2$-cover $\pi : \widetilde{\mathcal{G}} \to \mathcal{G}$ does not admit a splitting. Equivalently, the cocycle ω is not a coboundary on $\mathcal{G}$. (If $\mathsf{Sys} = \emptyset$, the claim is trivial.)*

Proof. If Sys is empty there is nothing to prove. Otherwise fix $r \in \mathsf{Sys}$, take $t = e_T$ and $u = 1$, and set $p = \mathrm{sqLoop}(t, u, r)$. If a splitting existed, Lemma .10 would imply $\omega(p) = 0$ for every loop. But Lemma .8 gives $\omega(p) = 1$, a contradiction. □

Proposition .1 (Two-way corner-path witness of the splitting obstruction)**.** *Fix any $t \in \mathsf{T}$, $u \in \mathbb{N}$, and $r \in \mathsf{Sys}$, and consider the two parallel morphisms*

$$p_{\mathrm{diag}} := \mathrm{toCorner}_{\mathrm{diag}}(t, u, r), \qquad p_{\mathrm{bridge}} := \mathrm{toCorner}_{\mathrm{bridge}}(t, u, r)$$

from r to $T_t(R_u(r))$ in $\mathcal{G}$. Then $\omega(p_{\mathrm{diag}}) = 0$ and $\omega(p_{\mathrm{bridge}}) = 1$. Consequently, no splitting of $\widetilde{\mathcal{G}} \to \mathcal{G}$ exists (as in Theorem .2); equivalently, the obstruction is already witnessed at this two-way corner configuration.

Proof. By definition, ω vanishes on time and rg and equals 0 on sq(diag) and 1 on sq(bridge), hence the two values are 0 and 1.

If a splitting existed, Lemma .10 would give $\sigma : \mathsf{Sys} \to \mathbb{Z}_2$ with $\omega(p) = \sigma(\mathrm{src}(p)) + \sigma(\mathrm{tgt}(p))$. Since p_{diag} and p_{bridge} have the same source and target, this would force $\omega(p_{\mathrm{diag}}) = \omega(p_{\mathrm{bridge}})$, contradiction. □

Corollary .3 (Three-checkpoint loop witness)**.** *Assume there exist a record state $r \in \mathsf{Sys}$, a time step $t \in \mathsf{T}$, and an RG step $u \in \mathbb{N}$ such that both procedures close at the record level:*

$$R_u(T_t(r)) = r \qquad \text{and} \qquad T_t(R_u(r)) = r.$$

(Equivalently, it suffices to assume one of these equalities once record-level interchange $T_t \circ R_u = R_u \circ T_t$ has been derived.)

Define the intermediate record checkpoints

$$A := r, \qquad B := T_t(r), \qquad C := R_u(r).$$

Then the two corner procedures of Proposition .1,

$$p_{\mathrm{diag}},\ p_{\mathrm{bridge}} :\ A \longrightarrow T_t(R_u(A)) = A,$$

are based loops at A whose $\mathbb{Z}_2$-labels differ:

$$\omega(p_{\mathrm{diag}}) = 0, \qquad \omega(p_{\mathrm{bridge}}) = 1.$$

Equivalently, even when the start and end coincide (three checkpoints), the distinction between "forgetting after" and "forgetting before" carries an intrinsic one-bit holonomy that cannot be reduced to a state-label on records.

Proof. Immediate from Proposition .1 after substituting the assumption $T_t(R_u(r)) = r$, which turns both parallel morphisms into loops based at r. □

Proposition .2 (Collapsed presentation splits)**.** *Let $\mathcal{G}^{\mathrm{coll}}$ be the quotient of $\mathcal{G}$ obtained by adding the relations*

$$\mathrm{sq}(\mathrm{diag}; t, u, r) \approx_5 \mathrm{sq}(\mathrm{bridge}; t, u, r) \qquad (t, u, r),$$

equivalently, identifying the two corner paths for each square. Then any functor $\mathcal{G}^{\mathrm{coll}} \to B\mathbb{Z}_2$ is trivial, and the induced $\mathbb{Z}_2$-cover splits.

Proof. The added relations force the images of the two square generators to agree, so the assignment $\omega(\mathrm{sq}(\mathrm{diag})) = 0$, $\omega(\mathrm{sq}(\mathrm{bridge})) = 1$ cannot descend. The only parity functor on $\mathcal{G}^{\mathrm{coll}}$ is therefore the trivial one, which is a coboundary (take $\sigma \equiv 0$), so the cover splits. □

Remark .6 (Tag collapse vs obstruction)**.** Theorem .2 concerns the non-collapsed pair $(\mathcal{G}, \omega)$. If one collapses the filler tags, one changes the input to $(\mathcal{G}^{\mathrm{coll}}, \omega_{\mathrm{triv}})$ as in Proposition .2. The semantic package in Section shows that such a collapse is incompatible with the intended interpretation on Diff: the detector supplies a non-fixed point of flip, so the induced ω-action is genuinely nontrivial.

DIFFERENCE OBJECT, FLIP, AND THE ω-ACTION

This section uses only Assumptions **(A1)**–**(A6)**.

Definition .10 (Set-level cofiber)**.** For a map $f : A \to B$, define $\mathrm{Cofib}(f)$ to be the quotient of $B \sqcup \{*\}$ by the relation that identifies all points $f(a)$ with the basepoint $*$. Let $\iota : B \to \mathrm{Cofib}(f)$ denote the induced map.

Define

$$\mathsf{Diff} := \mathrm{Cofib}(\mathrm{diag}), \qquad \iota : \mathsf{SK} \to \mathsf{Diff}.$$

Lemma .11 (Cofiber universal property)**.** *Let $f : A \to B$ and let X be a set. If $\phi : B \to X$ is constant on $\mathrm{im}(f)$, i.e. $\phi(f(a)) = x_0$ for some $x_0 \in X$ and all $a \in A$, then there exists a unique $\overline{\phi} : \mathrm{Cofib}(f) \to X$ with $\overline{\phi}(\iota(b)) = \phi(b)$ and $\overline{\phi}(*) = x_0$.*

Proof. Define $\overline{\phi}$ on representatives by $\overline{\phi}(\iota(b)) := \phi(b)$ and $\overline{\phi}(*) := x_0$. The hypothesis ensures this respects the identifying relation, hence descends to the quotient; uniqueness is immediate. □

Definition .11 (Flip involution)**.** The commuting square $\mathrm{swap} \circ \mathrm{diag} = \mathrm{diag}$ induces an endomorphism of the cofiber,

$$\mathrm{flip} : \mathsf{Diff} \to \mathsf{Diff}, \qquad \mathrm{flip}(\iota(x)) := \iota(\mathrm{swap}\, x), \qquad \mathrm{flip}(*) := *.$$

Then $\mathrm{flip}^2 = \mathrm{id}_{\mathsf{Diff}}$.

Lemma .12 (Canonical two fillers)**.** *Define* $\mathrm{toggle}(\mathrm{diag}) = \mathrm{bridge}$ *and* $\mathrm{toggle}(\mathrm{bridge}) = \mathrm{diag}$*, and define the filler action on* Diff *by*

$$\mathrm{actFill}(\mathrm{diag}) = \mathrm{id}_{\mathsf{Diff}}, \qquad \mathrm{actFill}(\mathrm{bridge}) = \mathrm{flip}\,.$$

Formally, actFill *is defined by composing the tag parity* $\mathrm{fillParity} : \{\mathrm{diag}, \mathrm{bridge}\} \to \{\mathrm{false}, \mathrm{true}\}$ *with the Bool-action on* Diff *induced by* flip*. Then for all tags f and all $x \in \mathsf{Diff}$,*

$$\mathrm{actFill}(\mathrm{toggle}\, f)(x) = \mathrm{flip}(\mathrm{actFill}(f)(x)), \qquad \mathrm{toggle}(\mathrm{toggle}\, f) = f.$$

In particular, the orbit generated by flip *is contained in* $\{\mathrm{id}, \mathrm{flip}\}$ *(so there are no further canonical choices; these two may coincide if* $\mathrm{flip} = \mathrm{id}$*).*

Definition .12 (ω-action on Diff)**.** The involution flip defines a $\mathbb{Z}_2$-action by $0 \cdot y := y$ and $1 \cdot y := \mathrm{flip}(y)$. For each morphism p in $\mathcal{G}$, define

$$\mathrm{act}_\omega(p)(y) := \omega(p) \cdot y.$$

Lemma .13 (Square-loop acts by flip)**.** *For all t, u, r and $y \in \mathsf{Diff}$,*

$$\mathrm{act}_\omega(\mathrm{sqLoop}(t, u, r))(y) = \mathrm{flip}(y).$$

Proof. By Lemma .8, $\omega(\mathrm{sqLoop}(t, u, r)) = 1$, and by definition of the $\mathbb{Z}_2$-action, $1 \cdot y = \mathrm{flip}(y)$. □

SEMANTIC NONTRIVIALITY FROM THE DERIVED DETECTOR

From here we use Assumption **(A7)**.

Lemma .14 (Derived detector gives a non-fixed point in Diff)**.** *Let* $\mathrm{obs} := \mathrm{obs}_{ab}$ *be the observable produced by Lemma .3. Then there exists $y \in \mathsf{Diff}$ with* $\mathrm{flip}(y) \neq y$.

Proof. By Lemma .2(i), obs is constant on diagonal record states, so Lemma .11 yields a unique factorization $\overline{\mathrm{obs}} : \mathsf{Diff} \to \mathbb{R}$ with $\overline{\mathrm{obs}}(\iota(x)) = \mathrm{obs}(x)$. Choose $x \in \mathsf{SK}$ such that $\mathrm{obs}(\mathrm{swap}\, x) \neq \mathrm{obs}(x)$ (exists by Lemma .3 and swap-oddness). Set $y := \iota(x)$. If $\mathrm{flip}(y) = y$, then applying $\overline{\mathrm{obs}}$ gives $\mathrm{obs}(\mathrm{swap}\, x) = \mathrm{obs}(x)$, contradiction. Hence $\mathrm{flip}(y) \neq y$. □

Corollary .4 (Distinguishability of canonical fillers)**.** *Under Assumption **(A7)**, there exists $y \in \mathsf{Diff}$ such that*

$$\mathrm{actFill}(\mathrm{bridge})(y) \neq \mathrm{actFill}(\mathrm{diag})(y).$$

Proof. By Lemma .14 choose y with $\mathrm{flip}(y) \neq y$. By Lemma .12, $\mathrm{actFill}(\mathrm{diag}) = \mathrm{id}$ and $\mathrm{actFill}(\mathrm{bridge}) = \mathrm{flip}$, hence the two values differ at y. □

Theorem .3 (Semantic nontriviality (detectable holonomy))**.** *Under Assumption **(A7)**, the ω-action on* Diff *is nontrivial: there exists $y \in \mathsf{Diff}$ with* $\mathrm{flip}(y) \neq y$*, and the square-loop acts nontrivially on* Diff*. In particular, the canonical fillers are distinguishable.*

Proof. Lemma .14 gives a non-fixed point of flip, and Lemma .13 identifies the ω-action of sqLoop with flip. Distinguishability follows from Corollary .4. □

Remark .7 (Why Diff is essential for semantics)**.** The oddness $\omega(\mathrm{sqLoop}) = 1$ is already a cohomological obstruction to splitting (Theorem .2); Diff is *not* needed for that. What Diff provides is a concrete $\mathbb{Z}_2$-action in which ω is realized as monodromy. The detector supplies a non-fixed point of flip, showing that this monodromy is genuinely nontrivial and the filler tags are operationally distinguishable.

Theorem .4 (Limited Scope Theorem)**.** *Under Assumptions **(A1)–(A7)**, the following hold:*

(1) Cohomological package (uses (A1)–(A6)). *The bicomplex identities $Q_t^2 = 0$, $Q_u^2 = 0$, and $Q_tQ_u + Q_uQ_t = 0$ (Theorem .1) hold; the presented groupoid $\mathcal{G}$ and cocycle ω are defined with $\omega(\mathrm{sqLoop}) = 1$ (Lemma .8); the $\mathbb{Z}_2$-cover splits iff ω is a coboundary (Lemma .10); the cover does not split (Theorem .2); and for any abelian group G the induced ω-action satisfies $\mathrm{act}_\omega^G(\mathrm{sqLoop}(t,u,r))(x) = -x$ (Corollary .1).*

(2) Semantic package (uses (A7)). *The ω-action on* Diff *is defined and* sqLoop *acts by* flip *(Lemma .13); the detector gives a non-fixed point of* flip *and hence distinguishable canonical fillers (Theorem .3).*

Proof. Item (1) collects the cited cohomological results; item (2) is the semantic nontriviality conclusion. □

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